

TLGM 120M 1024-Context vs MiniMind 120M

Local benchmark/report generated from trained checkpoints on RTX 4060 Ti 16GB

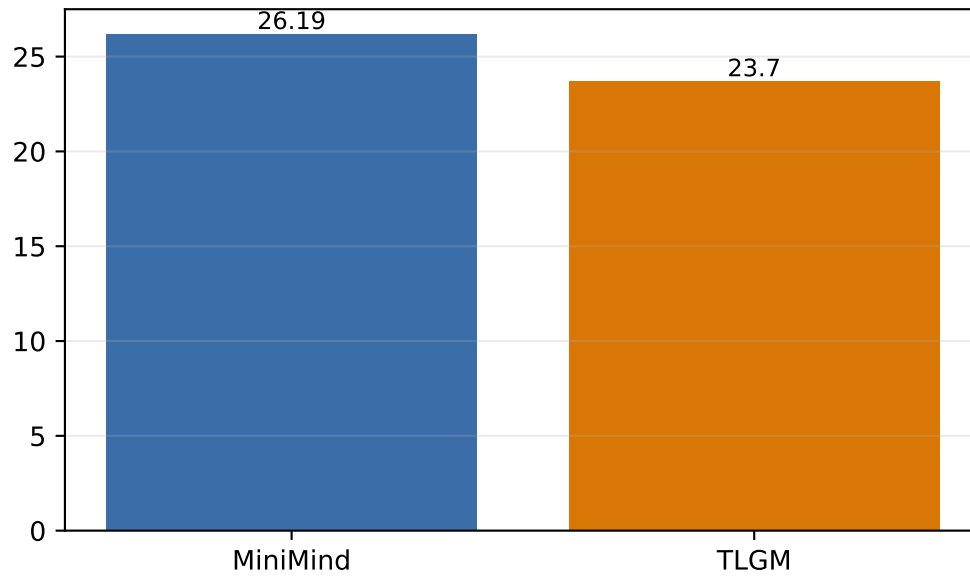
Generated: 2026-07-18

Executive summary

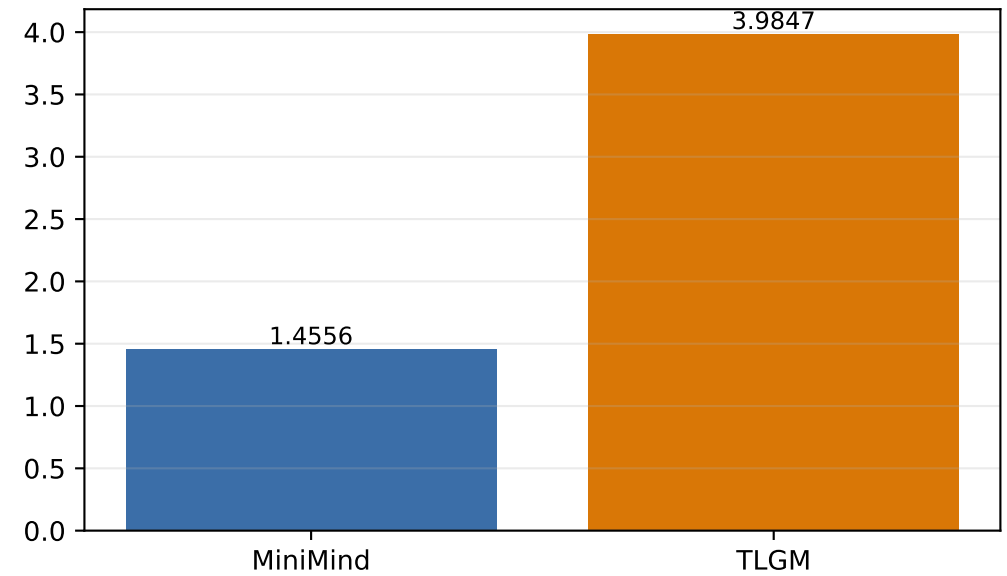
TLGM 1024ctx is faster in the measured inference benchmark and successfully supports 1024 context, but MiniMind remains the stronger language model. MiniMind's final pretraining loss is much lower, and its answers are generally richer. TLGM is a promising custom architecture baseline, but it needs architecture improvements beyond more SFT to compete with standard attention.

Key Quantitative Comparison

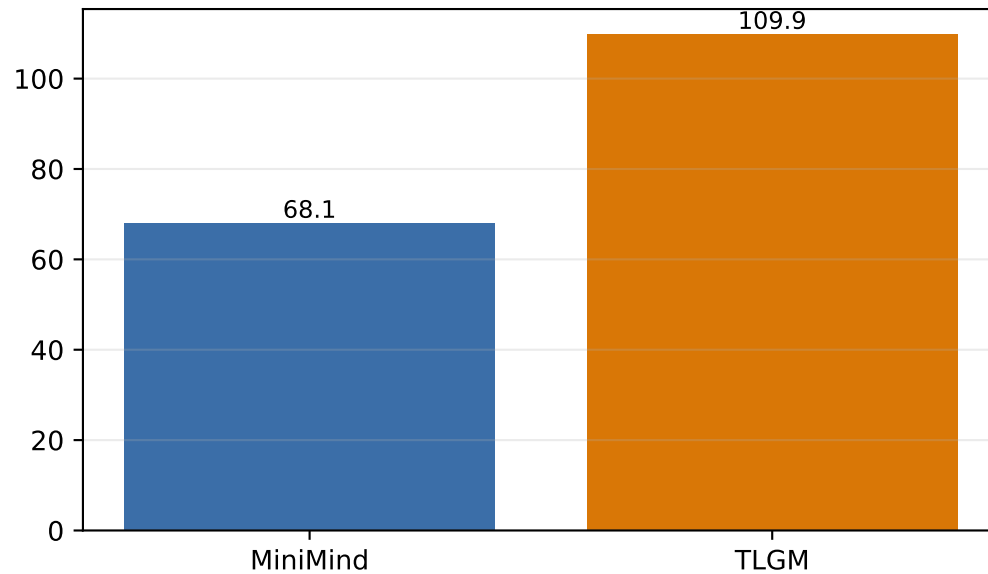
Pretrain Time (hours)



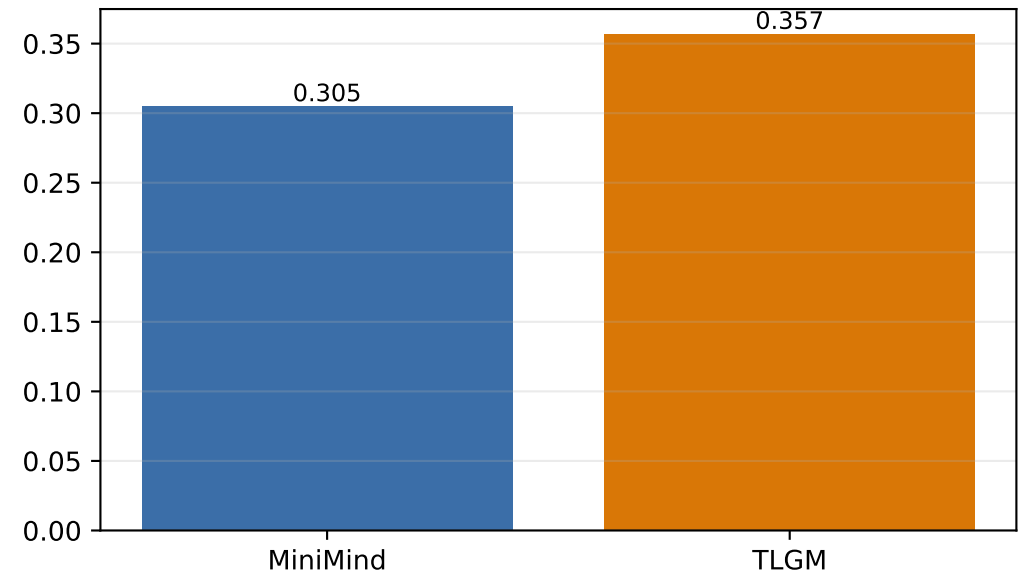
Final Pretrain Loss



1024 Prompt Generation (tok/s)



1024 Prompt Generation Memory (GB)



1. Compared Checkpoints

MiniMind 120M normal Transformer checkpoint:

C:\Users\ADMIN\minimind\minimind_120m_tiny_chatgpt\checkpoints\sft_120m_960_final.pth

TLGM 120M 1024-context custom non-Transformer checkpoint:

C:\Users\ADMIN\minimind\tlgm_120m_1024ctx\checkpoints\tlgm_120m_1024ctx_sft_polished.pt

Both models were trained from random initialization. Neither uses pretrained Qwen/Llama weights.

2. Architecture Summary

MiniMind 120M is a normal decoder-only Transformer. It uses self-attention, grouped-query attention, RoPE positional encoding, RMSNorm, feed-forward layers, tied embeddings, and generation. It is implemented through the official MiniMind model code.

TLGM 120M 1024ctx is a custom non-Transformer language model. It does not use self-attention, QKV projections, Mamba, RNN, LSTM, or GRU. It uses token embeddings, learned position embeddings, a shared per-token encoder, and a causal global mixing network. Sequence mixing is done with learned lower-triangular position-mixing matrices.

TLGM 1024ctx changed the older 256-context TLGM design by increasing context_length to 1024 and reducing global blocks to 16 so that the parameter count stays close to 120M.

3. Model Configuration

Metric	MiniMind 120M	TLGM 120M 1024ctx
Architecture	Transformer decoder	Causal sequence mixer
Parameters	122.29M	121.00M
Context length	1024	1024
Vocabulary size	6400	8192
Hidden/model dim	960	768
Layers/blocks	10 Transformer layers	16 global mixer blocks
Attention/QKV	Yes	No
KV cache	Yes	No
Tokenizer	MiniMind tokenizer	Scratch ByteLevel BPE

4. Training Summary

Metric	MiniMind 120M	TLGM 120M 1024ctx	Difference
Pretrain tokens	2.000B	2.000B	Same
Pretrain steps	30,518	30,518	Same
Tokens per step	65,536	65,536	Same
Pretrain time	26.19 h	23.70 h	TLGM 9.5% faster
Pretrain throughput	~21.2k tok/s	~23.4k tok/s	TLGM 10.5% faster
Final pretrain loss	1.4556	3.9847	TLGM 2.74x higher
SFT final/polish loss	0.0597	0.00005	Not directly comparable

5. Inference Memory - Forward / Prefill

Measured peak CUDA allocation, batch=1, FP16, random initialized models. Values are architecture/runtime measurements, not intelligence measurements.

Seq len	MiniMind	TLGM 1024ctx	TLGM more memory
128	264.5 MB / 0.258 GB	334.8 MB / 0.327 GB	+26.6%
512	274.2 MB / 0.268 GB	342.4 MB / 0.334 GB	+24.9%
1024	286.4 MB / 0.280 GB	351.9 MB / 0.344 GB	+22.9%

6. Inference Memory - 50 Token Generation

Measured peak CUDA allocation while generating 50 tokens, batch=1, FP16.

Prompt len	MiniMind	TLGM 1024ctx	TLGM more memory
128	262.8 MB / 0.257 GB	335.2 MB / 0.327 GB	+27.5%
512	283.5 MB / 0.277 GB	346.6 MB / 0.338 GB	+22.3%
1024	312.3 MB / 0.305 GB	365.3 MB / 0.357 GB	+17.0%

7. Speed - Forward / Prefill

Seq len	MiniMind	TLGM 1024ctx	TLGM speedup
128	12.11 ms	6.99 ms	1.73x faster
512	10.00 ms	6.29 ms	1.59x faster
1024	12.91 ms	8.41 ms	1.53x faster

8. Speed - 50 Token Generation

Prompt len	MiniMind	TLGM 1024ctx	TLGM speedup
128	70.0 tok/s	144.8 tok/s	2.07x faster
512	67.1 tok/s	153.1 tok/s	2.28x faster
1024	68.1 tok/s	109.9 tok/s	1.61x faster

9. Disk / Checkpoint Size

TLGM's final checkpoint is large because it stores full training state, including optimizer and scaler. A model-only FP16 TLGM checkpoint should be around 0.25 GB, similar to MiniMind.

Metric	MiniMind 120M	TLGM 120M 1024ctx	Note
Params	122.29M	121.00M	TLGM 1.1% smaller
Full training checkpoint	1.235 GB	1.223 GB	Similar
Final model-only checkpoint	0.257 GB	Not currently saved model-only	TLGM stores optimizer state

10. Simple Prompt Evaluation

Both models answered the basic tests correctly after SFT/polish:

hello -> Hello! How can I help you?

what is 1+5 -> 1+5 is 6.

capital of France -> The capital of France is Paris.

cat/dog/chicken -> correct simple definitions.

ghost -> stated that ghosts are story/spirit concepts and no reliable scientific evidence exists.

fake moon chess championship -> correctly said it does not know and should not invent a

Observed qualitative difference: MiniMind usually gives longer and richer answers. TLGM shorter, cleaner answers that look strongly shaped by the polish SFT examples.

11. MiniMind 120M Pros and Cons

Pros:

- Stronger language-modeling architecture.
- Much better pretraining loss after the same 2B token budget.
- Better natural answer detail.
- Uses attention, RoPE, grouped-query attention, and KV cache.
- Better suited for factual recall, instruction following, longer chat, and future improvements.
- Smaller final model-only checkpoint is already available.

Cons:

- Slower pretraining on this hardware.
- Slower measured generation than TLGM 1024ctx.
- Transformer internals are more complex.
- Less experimental/custom if the goal is to test non-attention architectures.

12. TLGM 120M 1024ctx Pros and Cons

Pros:

- Custom non-Transformer architecture works and now supports 1024 context.
- Faster measured prefill and generation than MiniMind on this RTX 4060 Ti test.
- Slightly fewer parameters.
- Trained 2B tokens faster than MiniMind.
- Stable on simple chat after masked SFT, repair SFT, and polish SFT.

Cons:

- Pretraining loss is much worse: 3.9847 vs 1.4556.
- Uses 17% to 28% more measured inference memory depending on prompt length/test mode.
- No KV-cache.
- Lower-triangular position mixer is less content-adaptive than attention.
- More likely to memorize SFT examples.
- Weaker expectation for unseen facts, reasoning, and complex instructions.
- Final checkpoint is large until a model-only export is created.

13. Final Verdict

TLGM 120M 1024ctx is a successful experimental result: it trains from scratch, reaches context, chats coherently on simple prompts, and runs faster than MiniMind in the measured inference benchmark.

However, MiniMind 120M remains the stronger tiny ChatGPT candidate. The main evidence is pretraining loss gap: MiniMind reached 1.4556 while TLGM reached 3.9847 on the same 2B-budget. That means MiniMind learned the general text distribution much better.

Recommended direction:

- For best practical tiny assistant: keep improving MiniMind with better SFT/preference
- For custom architecture research: continue TLGM, but focus on architecture changes, not more SFT. The mixer needs more content-adaptive behavior to close the gap with attention